

Rohan Gupta

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Driven and excited about opportunities in AI/ML

EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA

BS Computer Science, Finance Minor

MS Computer Science, emphasis in AI/ML & Robotics

GPA: 3.73/4

Relevant Activities: Undergraduate TA for introductory Artificial Intelligence and Operating Systems courses

Los Angeles, CA

Aug 2020 - May 2024

Jan 2024 to May 2025

RELEVANT EXPERIENCE

SYNOPSYS | Senior Machine Learning Engineer

R&D Machine Learning Intern

July 2025 – Present

June 2024 – Feb 2025

- Owned research and full-stack implementation (~70K LoC across ~100K-line codebase) of a transformer-based model for chip design optimization; achieved Pareto-dominant power-timing tradeoffs that beat baseline and expert-designed configurations for placement and routing
- Adapted LLM training approaches to EDA; designed novel custom GPT-inspired decoder model with multi-source embedding composition, EDA feature encodings, and multi-task heads (logits, EDA score prediction, etc.) that generates recommendation tokens for chip designs
- Engineered robust production-grade ML infrastructure using Docker, Hydra, and MLFlow; implemented and optimized distributed training via PyTorch FSDP/DDP with custom logging for activation/gradient metrics across 100+ training runs
- Implemented and evaluated supervised pre-training using Bradley-Terry margin-ranking with 3-stage curriculum learning and offline DPO; developed online RL finetuning loop (primarily PPO) using EDA tool calls as the environment and reward function
- Developed constrained autoregressive decoding inference (beam search, greedy, and sampling) conditioned on optimization intent (timing, power, leakage) and chip design stage (placement, CTS, routing); built closed-loop flow using EDA tool outputs on inference outcomes for online finetuning
- Architected data collection and featurization pipeline using EDA tools, proprietary design databases, YAML-based configurable feature extraction and selection, and custom tokenizers; built training datasets using Pydantic-typed data models and optimized PyTorch training memory management with data stores and caches
- Built a comprehensive full-stack Python and React model analytics dashboard for analysis of general model training dynamics and EDA-specific performance metrics
- Led team efforts in mechanistic interpretability by building a sparse autoencoder training pipeline to analyze internal model representations
- Contributed to secondary projects in netlist tokenization and foundation models, JAX and CUDA-based speed ups of differentiable optimization in chip placement algorithms, and tool-calling LLM agents for flagship Fusion Compiler product
- Drove high-impact features within lean, 3-4-person team with full ownership of model development and ML expertise

USC VITERBI SCHOOL OF ENGINEERING | AI/ML Research Assistant & Engineer

May 2023 – July 2025

- Won university research award to collaborate with Professor Jesse Thomason in the GLAMOR Lab on robotics, computer vision, and NLP research for Vision-and-Language Navigation and Sim2Real
- Collected, trained and fine-tuned language-conditioned policies (ACT, Cross Modal Attention, Diffusion, etc.) on multimodal robot data for mobile manipulators and navigation robots using reinforcement learning for finetuning and supervised learning, specifically behavior-cloning
- Contributed to a heterogeneous multi-robot coordination codebase using ROS1/ROS2 and diverse hardware libraries (Toyota HSR, Trossen, Stretch)
- Worked on top of public repos like TeleMoMa and LeRobot and fixed bugs in them as well as added ROS1 to ROS2 upgrades for them
- Built a sample-efficient evaluation framework for a physical robot using perplexity and distance-to-goal as success metrics
- Researched how contrast sets can be applied to robot experiments to save time and experimenter effort — co-first author on both papers for this project that got accepted to a [CoRL 2023 Workshop](#) (Oral, Top 15% of papers) and [CoRL 2024](#) (38.2% acceptance rate)
- Built simulation data collection and policy training pipelines using MetaWorld for an efficient policy evaluation project that got accepted to [CoRL 2025](#).
- Current project “RobotFleet” is heterogenous multi-robot coordination using natural language LLMs and multimodal policies. Paper accepted to RSS 2025 Multi-Robot Systems Workshop and GitHub repository [open-sourced](#).

INTERNSHIPS AND OTHER PROJECTS | Co-founder, student developer, intern

- USC Investment Office** Jun – Dec 2022
Focused on software, automation, and time-series financial and data analysis projects to support running \$8B+ endowment.
- StreetFins** Oct 2017 – Nov 2022
Built and grew team of 60+ students for a financial education startup. Created the business plan, coded the website, and reached out successfully to high schools across 16 states. Hosted the “Finance Simplified” podcast which has over 70k listens with notable guests from industry, academia, and government. Won paid contracts with 2 larger companies to produce educational content.
- Classical Music Generation** Sep – Dec 2023
Used the music21 library and PyTorch to do supervised learning and train transformers, RNNs, and n-grams to generate classical music. Created a custom data transformation pipeline for musical compositions from 9 classical composers and trained models on that data.

SKILLS

- Languages:** Python, C/C++, Java, JavaScript, TypeScript, Angular, React, PHP, SQL
- ML/Data:** PyTorch, JAX, CUDA, Hydra, WandB, MLFlow, Vector DBs, LangChain, Stable Baselines
- Tools:** Docker, Linux, AWS, ROS1/2, Git, OpenCV